

PAYNESVILLE, MN, USA

WMO: 720367

Lat: 45.372N

Lon: 94.746W

Elev: 360

StdP: 97.08

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-24.4	-30.9	0.2	-26.8	-27.8	0.3	-23.9	11.1	-10.2	10.3	-9.1	2.2	280	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	31.1	21.9	29.1	21.1	27.8	20.1	24.3	28.8	23.1	27.1	22.0	26.0	4.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.3	27.1	21.9	17.4	26.1	20.8	16.2	24.5	75.3	28.5	70.3	27.3	66.1	26.0	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.8	DB	-31.3	33.7	3.0	2.2	-33.5	35.3	-35.2	36.6	-36.9	37.9	-39.1	39.5	(4)
(5)			WB	-31.5	25.9	2.9	1.1	-33.6	26.7	-35.3	27.3	-37.0	28.0	-39.1	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-11.2	-9.7	-1.1	6.1	13.5	18.7	21.3	19.6	15.6	7.7	-0.5	-8.0	(6)	
	DBStd	12.72	7.83	6.99	7.04	5.76	5.03	3.46	3.27	3.02	4.71	5.19	6.24	6.80	(7)	
	HDD10.0	2710	658	552	349	139	21	0	0	0	7	108	318	557	(8)	
	HDD18.3	4724	917	786	602	366	167	36	11	21	107	331	566	815	(9)	
	CDD10.0	1281	0	0	5	23	129	261	349	298	176	37	2	0	(10)	
	CDD18.3	253	0	0	0	0	16	47	101	61	26	2	0	0	(11)	
	CDH23.3	2371	0	0	0	7	204	421	971	511	231	26	0	0	(12)	
	CDH26.7	623	0	0	0	1	64	102	278	118	57	4	0	0	(13)	
(14) Wind	WSAvg	3.4	3.6	3.6	3.7	4.1	3.8	3.2	2.7	2.6	3.2	3.7	3.7	3.5	(14)	
(15) Precipitation	PrecAvg	764	20	20	42	72	84	123	99	94	82	59	37	19	(15)	
	PrecMax	975	51	51	89	179	171	203	188	192	208	123	101	42	(16)	
	PrecMin	595	1	3	14	21	22	51	22	9	13	19	1	8	(17)	
	PrecStd	116	15	13	21	39	37	39	42	40	48	30	32	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	10.5	20.0	24.0	32.2	31.5	33.8	31.4	31.0	27.0	17.4	7.7	(19)	
		MCWB	2.4	8.2	14.0	14.1	18.7	21.9	24.7	22.5	21.7	16.7	11.5	5.1	(20)	
	2%	DB	2.6	4.5	15.9	21.1	28.0	29.1	31.3	29.3	28.2	22.5	13.8	4.5	(21)	
		MCWB	1.2	2.5	11.4	12.1	17.8	20.5	22.9	22.2	20.3	14.8	8.6	2.5	(22)	
	5%	DB	1.2	2.2	12.0	18.4	25.2	27.5	29.3	27.8	26.2	18.9	11.0	2.4	(23)	
		MCWB	0.4	0.9	9.3	10.7	15.9	19.1	21.9	20.5	19.1	13.3	7.1	1.3	(24)	
	10%	DB	-0.2	0.1	8.2	15.9	22.7	25.9	27.8	26.3	23.8	16.2	7.8	1.0	(25)	
		MCWB	-0.6	-0.8	6.0	9.3	15.2	18.4	20.8	19.8	18.4	11.3	5.2	0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	8.3	15.6	15.2	22.1	24.6	26.4	24.8	24.2	19.5	13.3	5.7	(27)	
		MCDB	5.0	11.5	18.5	22.2	29.0	29.0	30.2	29.1	28.8	23.0	17.4	7.0	(28)	
	2%	WB	1.4	2.9	12.0	13.3	19.9	22.6	25.0	23.6	22.2	16.1	9.9	2.7	(29)	
		MCDB	2.4	4.2	13.9	18.6	25.3	26.8	29.7	27.6	25.5	20.3	12.1	4.0	(30)	
	5%	WB	0.4	0.8	8.7	11.7	17.9	20.8	23.6	22.4	20.8	14.0	7.6	1.3	(31)	
		MCDB	1.1	2.0	11.5	16.5	22.3	25.0	28.0	26.1	24.3	18.3	10.1	2.2	(32)	
	10%	WB	-0.9	-0.8	5.9	10.1	16.3	19.7	22.2	21.3	19.2	11.9	5.4	0.3	(33)	
		MCDB	-0.3	0.0	8.4	15.0	20.8	23.8	26.4	24.7	22.6	14.9	8.0	0.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.2	9.3	10.8	12.0	11.1	11.7	11.4	11.7	10.4	9.1	8.4	(35)	
		MCDBR	9.8	11.0	13.6	16.0	16.3	13.1	12.4	12.9	14.0	14.5	12.3	8.9	(36)	
	5% WB	MCWBR	8.4	9.1	8.9	8.6	7.5	6.2	5.9	6.4	6.9	8.0	8.0	7.6	(37)	
		MCWBR	9.3	10.4	12.3	12.9	12.6	11.1	10.8	10.9	11.3	12.2	10.9	8.2	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.260	0.277	0.308	0.363	0.391	0.387	0.385	0.398	0.360	0.312	0.281	0.255	(40)	
		taud	2.426	2.371	2.402	2.316	2.299	2.358	2.375	2.327	2.414	2.535	2.521	2.456	(41)	
	Edn at Noon	Edn at Noon	869	913	927	895	878	880	877	851	859	865	838	838	(42)	
		Edh at Noon	69	89	101	120	127	120	117	118	100	76	64	61	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.74	2.82	3.79	4.70	5.41	6.01	6.44	5.36	4.11	2.61	1.67	1.22	(44)	
	RadStd	RadStd	0.22	0.32	0.35	0.41	0.42	0.41	0.23	0.44	0.31	0.37	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page